

University of the Punjab

(Gujranwala Campus)

ARTIFICIAL INTELLIGENCE

LAB TASK 02

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BSIT (5th Semester)

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TASK 01

main.py | Visualize | Run ▶ | Output

```
1 # Python program to illustrate
2 # while loop
3 count = 0
4 while (count < 3):
5   count = count + 1
6   print("Hello Geek")
```

```
Hello Geek
Hello Geek
Hello Geek
```

TASK 02

main.py | Visualize | Run ▶ | Output

```
1 # Python program to illustrate
2 # Single statementwhile block
3 count = 0
4 while (count == 0): print("Hello Geek")
```

```
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
Hello Geek
```

TASK 03

```python
# Python program to illustrate
# Iterating over a list
print("List Iteration")
l = ["geeks", "for", "geeks"]
for i in l:
  print(i)

```

```
List Iteration
geeks
for
geeks
```

TASK 04

main.py Visualize Run Output

```python
# Iterating over a tuple (immutable)
print("\nTuple Iteration")
t = ("geeks", "for", "geeks")
for i in t:
  print(i)
```

```
Tuple Iteration
geeks
for
geeks
```

TASK 05

main.py Visualize Run Output

```python
# Iterating over a String
print("\nString Iteration")
s = "Geeks"
for i in s :
  print(i)
```

```
String Iteration
G
e
e
k
s
```

TASK 06

main.py | Visualize | Run ▶ | Output

```
# Python program to illustrate
# Iterating by index
list = ["geeks", "for", "geeks"]
for index in range(len(list)):
  print (list[index])
```

```
geeks
for
geeks
```

TASK 07

main.py | Visualize | Run ▶ | Output

```
# Prints all letters except 'e' and 's'
for letter in 'geeksforgeeks':
  if letter =='e' or letter =='s':
     continue

  print('Current Letter :',letter)
  var = 10
```

```
Current Letter : g
Current Letter : k
Current Letter : f
Current Letter : o
Current Letter : r
Current Letter : g
Current Letter : k
```

TASK 08

main.py | Visualize | Run ▶ | Output

```
for letter in 'geeksforgeeks':
  # break the loop as soon it sees 'e' # or 's'
  if letter == 'e' or letter == 's':
     break

  print ('Current Letter :', letter)
```

```
Current Letter : g
```

TASK 09

```
def my_function():
  print("Hello from a function")
my_function()
```

TASK 1O

```
def my_function(fname):
  print(fname + " Refsnes")
my_function("Emil")
my_function("Tobias")
my_function("Linus")
```

```
Emil Refsnes
Tobias Refsnes
Linus Refsnes
```

TASK 11

main.py Visualize Run ▶ Output

```
def my_function(country = "Norway" ):
  print("I am from " + country)
my_function("Sweden")
my_function("India")
my_function()
my_function("Brazil")
```

```
I am from Sweden
I am from India
I am from Norway
I am from Brazil
```

TASK 12

```
def my_function(food):
  for x in food:
    print(x)
fruits = ["apple", "banana", "cherry"]
my_function(fruits)
```

```
apple
banana
cherry
```

TASK 13

main.py Visualize Run ▶ Outpu

```
def my_function(x):
  return 5 * x
print(my_function(3))
print(my_function(5))
print(my_function(9))
```

```
15
25
45
```

TASK 14

main.py Visualize Run ▶ Output

```
def my_function(child3, child2, child1):
  print("The youngest child is " + child3)
my_function(child1 = "Emil", child2 = "Tobias", child3 = "Linus")
```

```
The youngest child is Linus
```

TASK 15

main.py | Visualize | Run ▶ | Output

```
1 class MyClass:
2   x = 5
3 p1 = MyClass()
4 print(p1.x)
```

```
5
```

TASK 16

main.py | Visualize | Run ▶ | Output

```
1 class Person:
2    def __init__(self, name, age):
3        self.name = name
4        self.age = age
5
6 p1 = Person ("John", 36)
7
8 print(p1.name)
9 print(p1.age)
```

```
John
36
```

TASK 17

main.py | Visualize | Run ▶ | Output

```
1 class Person:
2   def __init__ (self, name, age):
3     self.name = name
4     self.age = age
5   def myfunc(self):
6     print("Hello my name is " + self.name)
7
8 p1 = Person("John", 36)
9 p1.myfunc()
```

```
Hello my name is John
```